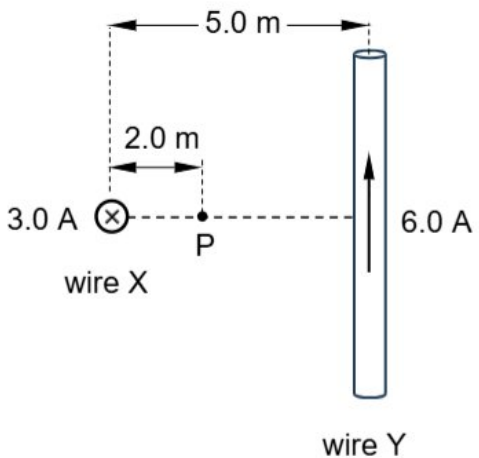
24 Two long, straight, current-carrying wires, X and Y are at a perpendicular distance 5.0 m apart. The current in X is 3.0 A and into the plane of the paper, while current in Y of 6.0 A is towards the top of the page.

Point P is on the line of shortest distance joining the two wires, at a distance 2.0 m away from wire X as shown.



What is the magnitude of the resultant magnetic flux density at point P?

A 3.5×10^{-12} T

B 5.0×10^{-7} T

C 6.3×10^{-7} T

D 7.0×10^{-7} T